Mixed−Type Patient Clustering Benchmark

A CSV−first comparison of eight mixed−data clustering strategies

400 patients | 100 mixed features | 4 known clusters

The report analyzes mock_patients.csv using distance−based, prototype−based, dimension−reduced, probabilistic, spectral, and consensus clustering. Dedicated heatmaps show the patient structure produced by every method.

Reproducible seed: 20260902

Executive summary

Headline results from the executed benchmark

10.9%	0.554	85.0%
Overall missing cells	Consensus ARI	Mean membership certainty

- The best external recovery was k−prototypes (ARI 0.615). The grand consensus achieved ARI 0.554 and subsample stability 0.838.
- Across MAR−style imputations, the consensus partitions had mean pairwise ARI 0.794. This isolates uncertainty associated with missing values from ordinary resampling instability.
- Directional MNAR scenarios reassigned between 0.0% and 0.0% of patients relative to the baseline scenario.
- No single internal index proves that 4 biological subtypes exist. Retain K=1 as a possibility and require clinical interpretability, resampling stability, and external replication.
- Known labels are used to score methods only. They are never passed to imputation, dimension reduction, or clustering.

Input data overview

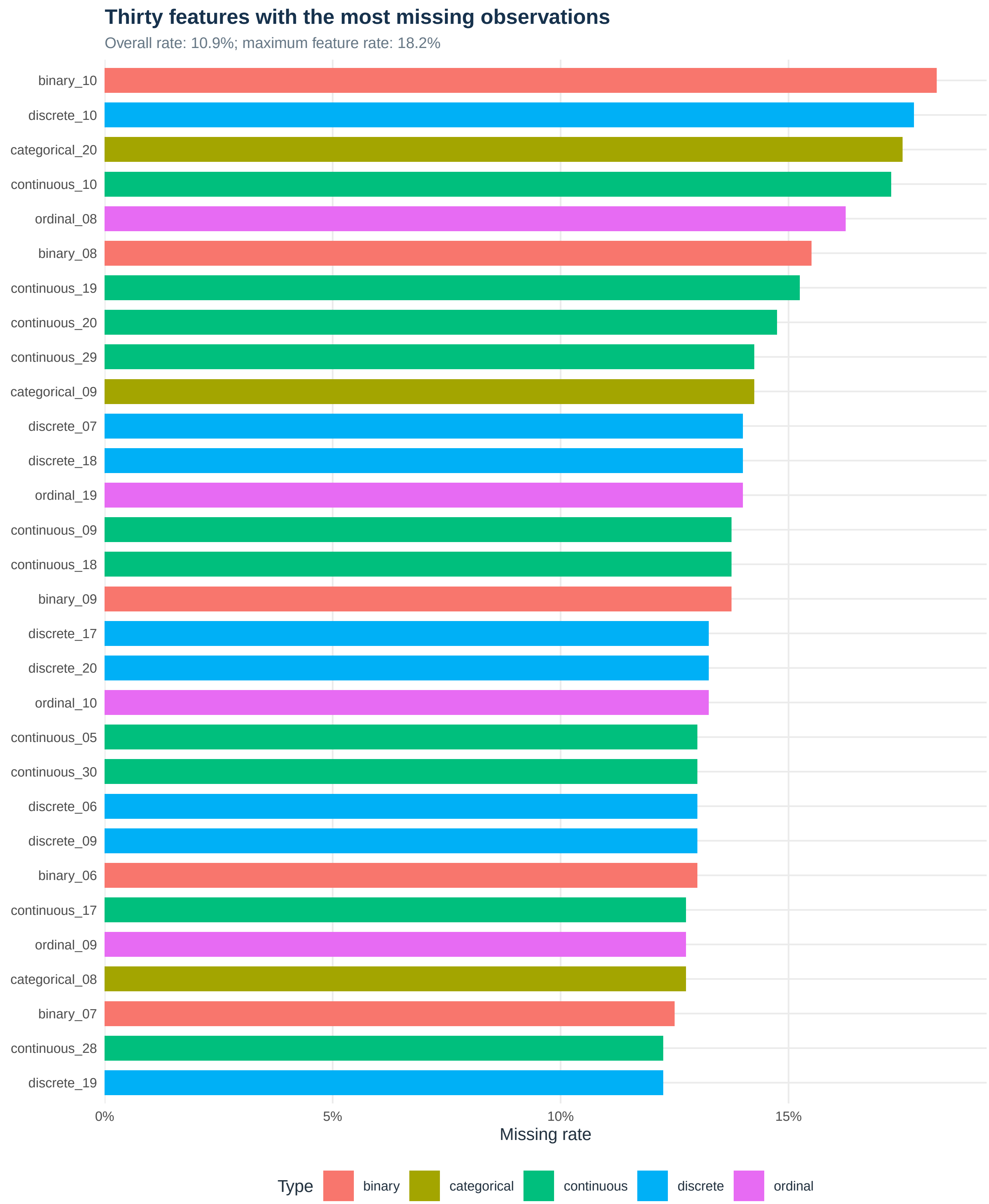
The feature dictionary controls types, domains, and excluded metadata

Feature type	Features	Mean missing
continuous	30	10.9%
discrete	20	11.0%
ordinal	20	10.7%
categorical	20	10.6%
binary	10	12.0%

- Input: mock_patients.csv. The analysis used 400 rows and 100 clustering features.
- Feature types were inferred from the CSV and saved to feature_dictionary_used.csv. Review ordinal variables and clinical domains before treating this as a final analysis.
- The dictionary contains 100 weighting domains. Each domain receives equal total Gower weight, so large feature blocks do not dominate solely by column count.
- Observed missingness is 10.9% overall. Stochastic mixed−type hot−deck imputations preserve observed categories and propagate assignment uncertainty.
- A truth column was supplied for external evaluation and excluded from all fitting steps.

Observed missingness

Feature−level rates; colors indicate declared measurement type



Approaches implemented

All methods use the same type dictionary and imputed datasets

Approach	Representation	Assignment	Key caveat
Gower + PAM	Weighted mixed dissimilarity	Medoid	Feature weighting
Gower + hierarchical	Weighted mixed dissimilarity	Average−linkage cut	Linkage sensitivity
k−prototypes	Standardized numeric + category modes	Prototype	Lambda and local minima
FAMD−style + k−means	Balanced mixed latent axes	Centroid	Two−stage approximation
Latent−embedding mixture	Balanced mixed latent axes	Posterior mode	Latent Gaussian assumption
Latent class/profile	Typed conditional distributions	Posterior mode	Conditional independence
Gower spectral	Gower affinity eigenvectors	Spectral centroid	Kernel bandwidth
Grand consensus	Co−assignment matrix	Consensus medoid	Can average weak answers

Implementation note: the FAMD−style embedding is a dependency−light PCAmix approximation. It standardizes numeric/ordinal variables and balances each nominal indicator block before PCA. Production work can replace it with FactoMineR::FAMD or PCAmixdata without changing the evaluation harness.

Method performance

Consensus per method across eight imputations; higher is better except runtime

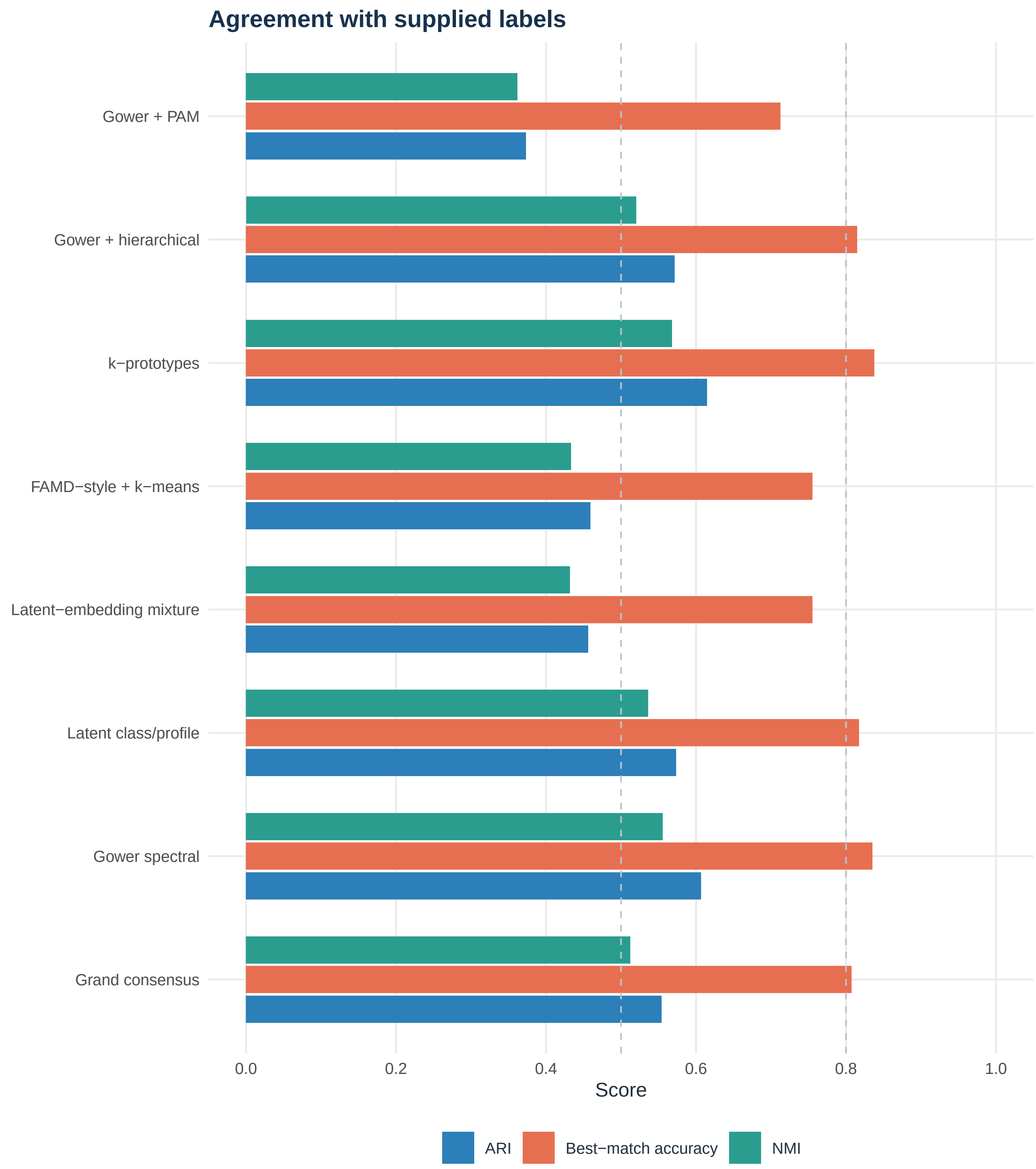
Method	ARI	NMI	Accuracy	Silhouette	MI stability	Subsample stability	Runtime (s)
Gower + PAM	0.373	0.362	71.2%	0.042	0.304	0.321	0.01
Gower + hierarchical	0.572	0.520	81.5%	0.052	0.347	0.408	0.00
k−prototypes	0.615	0.568	83.8%	0.055	0.828	0.863	3.46
FAMD−style + k−means	0.459	0.434	75.5%	0.050	0.799	0.814	0.01
Latent−embedding mixture	0.456	0.432	75.5%	0.048	0.780	0.751	0.03
Latent class/profile	0.574	0.536	81.8%	0.052	0.813	0.804	0.23
Gower spectral	0.607	0.556	83.5%	0.056	0.815	0.852	0.05
Grand consensus	0.554	0.512	80.8%	0.054	0.794	0.838	3.78

ARI and NMI compare clusters to the supplied truth labels. Accuracy is calculated after the best one−to−one label permutation. Silhouette uses the first completed−data Gower distance. MI stability is mean pairwise ARI across imputation−specific partitions. Subsample stability compares refitted 80% subsets with the full−data solution on the same patients.

Runtime values are indicative on this machine and omit report rendering. Shared preprocessing makes exact cross−method timing comparisons approximate.

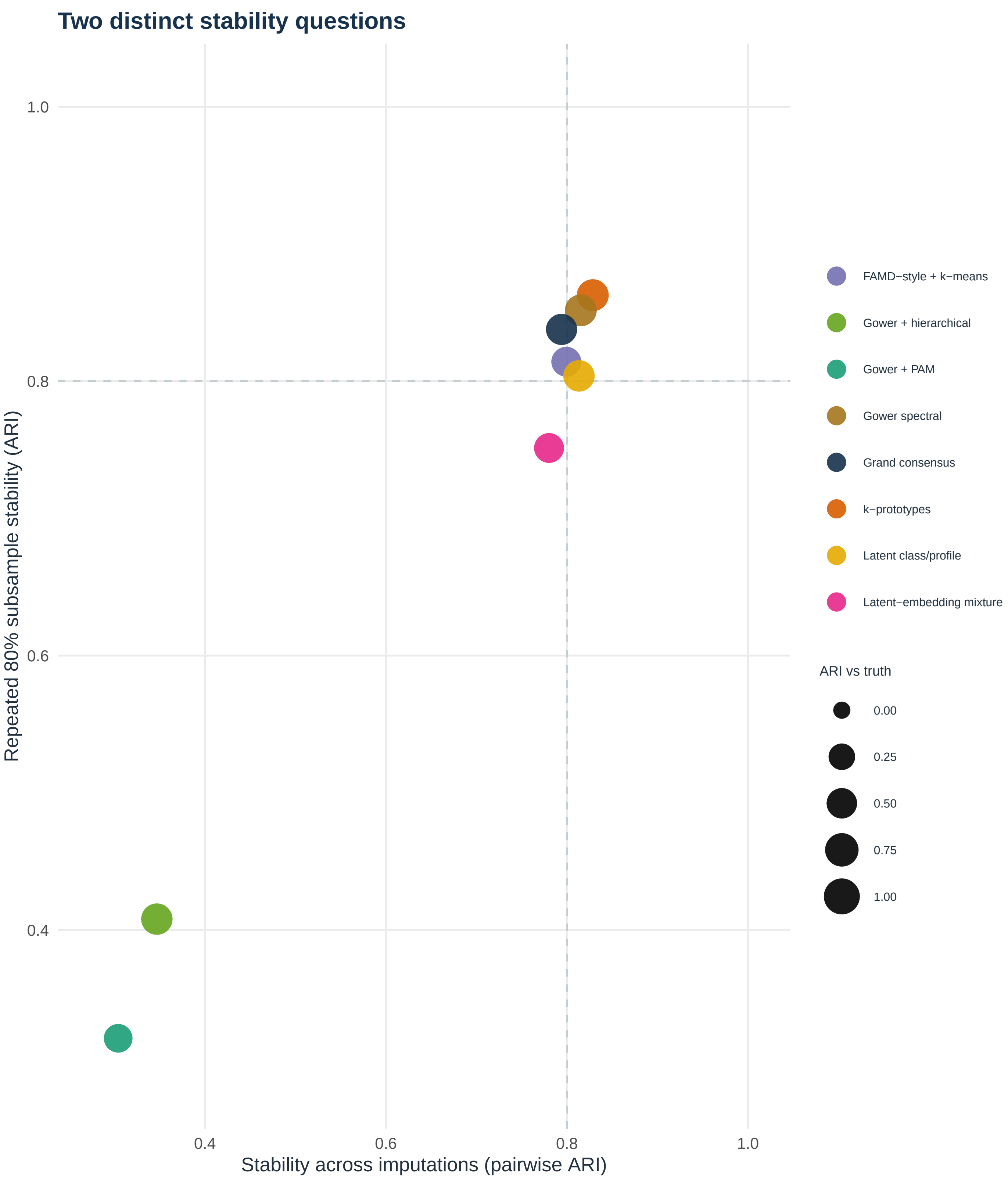
Recovery of supplied labels

External metrics are used only for evaluation



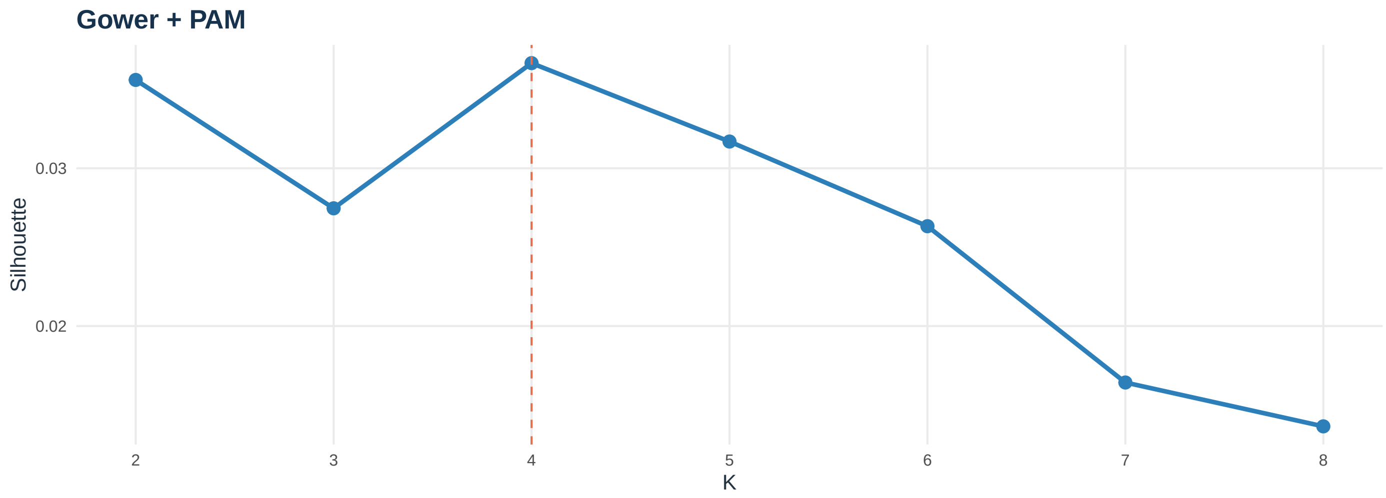
Stability and missing−data uncertainty

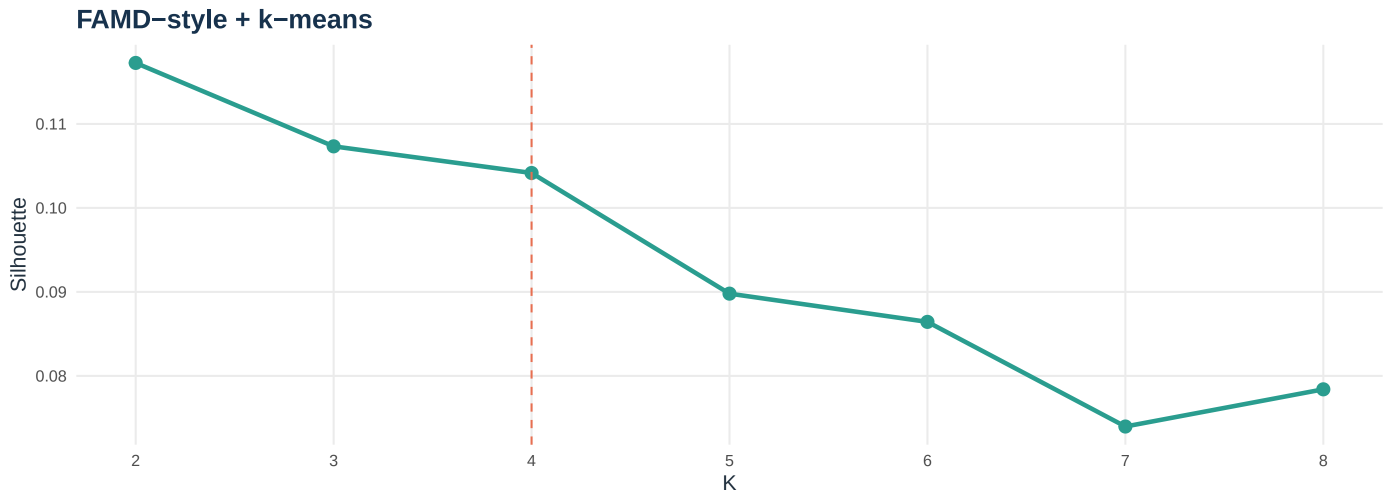
Strong recovery is useful only when assignments survive perturbation

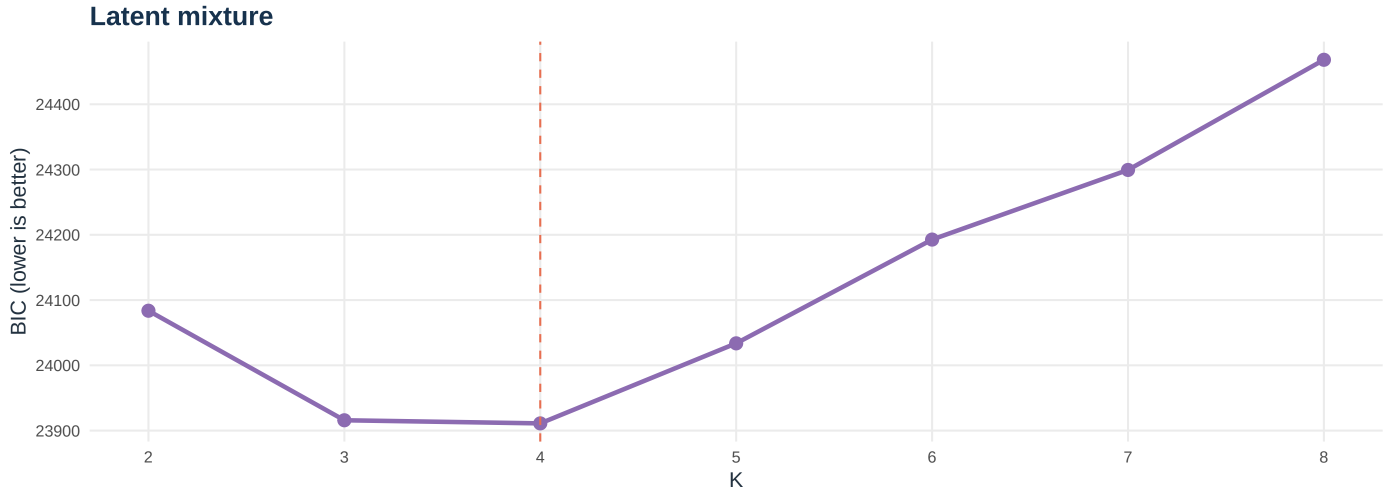


Choosing the number of clusters

No single criterion should determine K

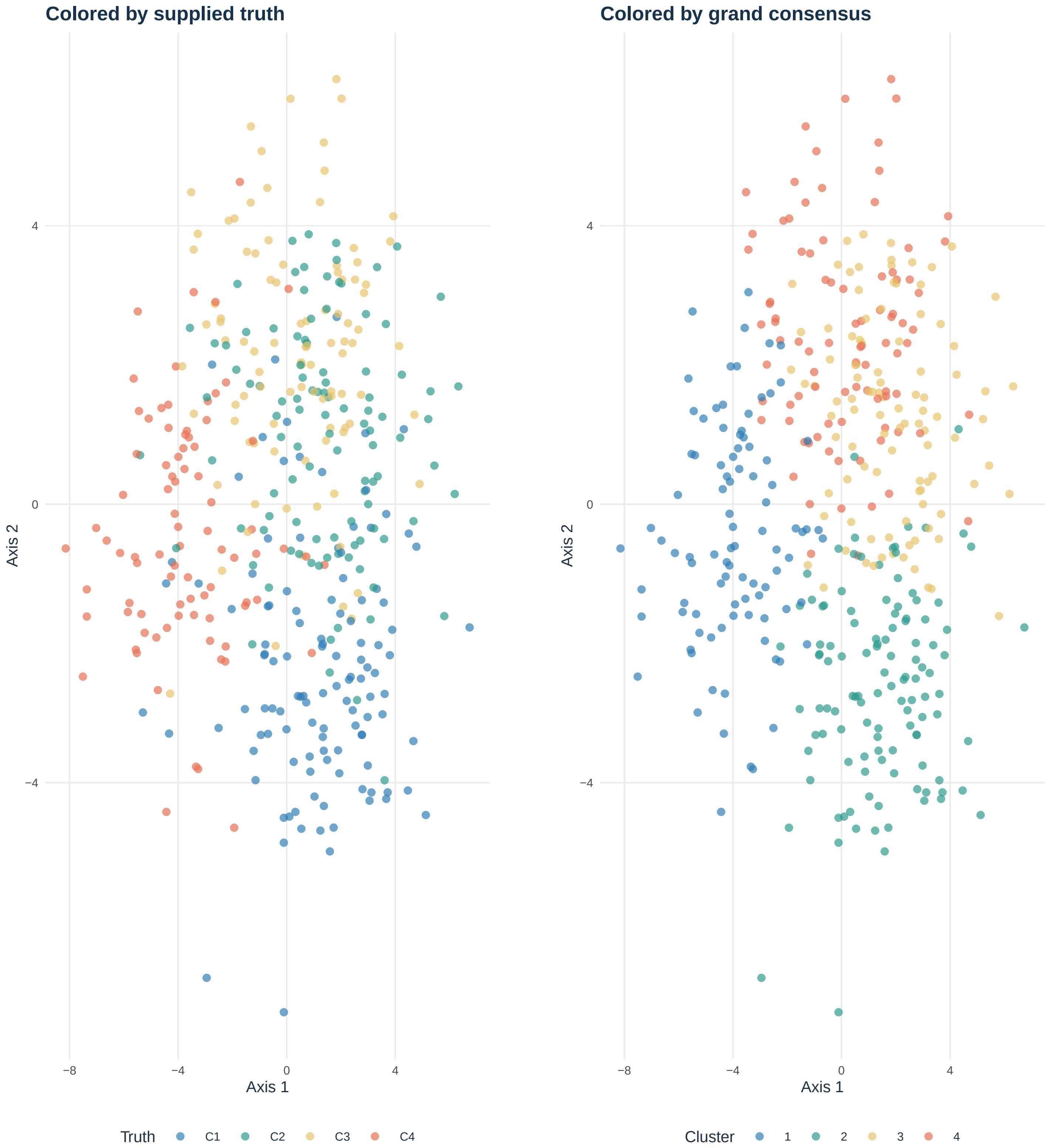






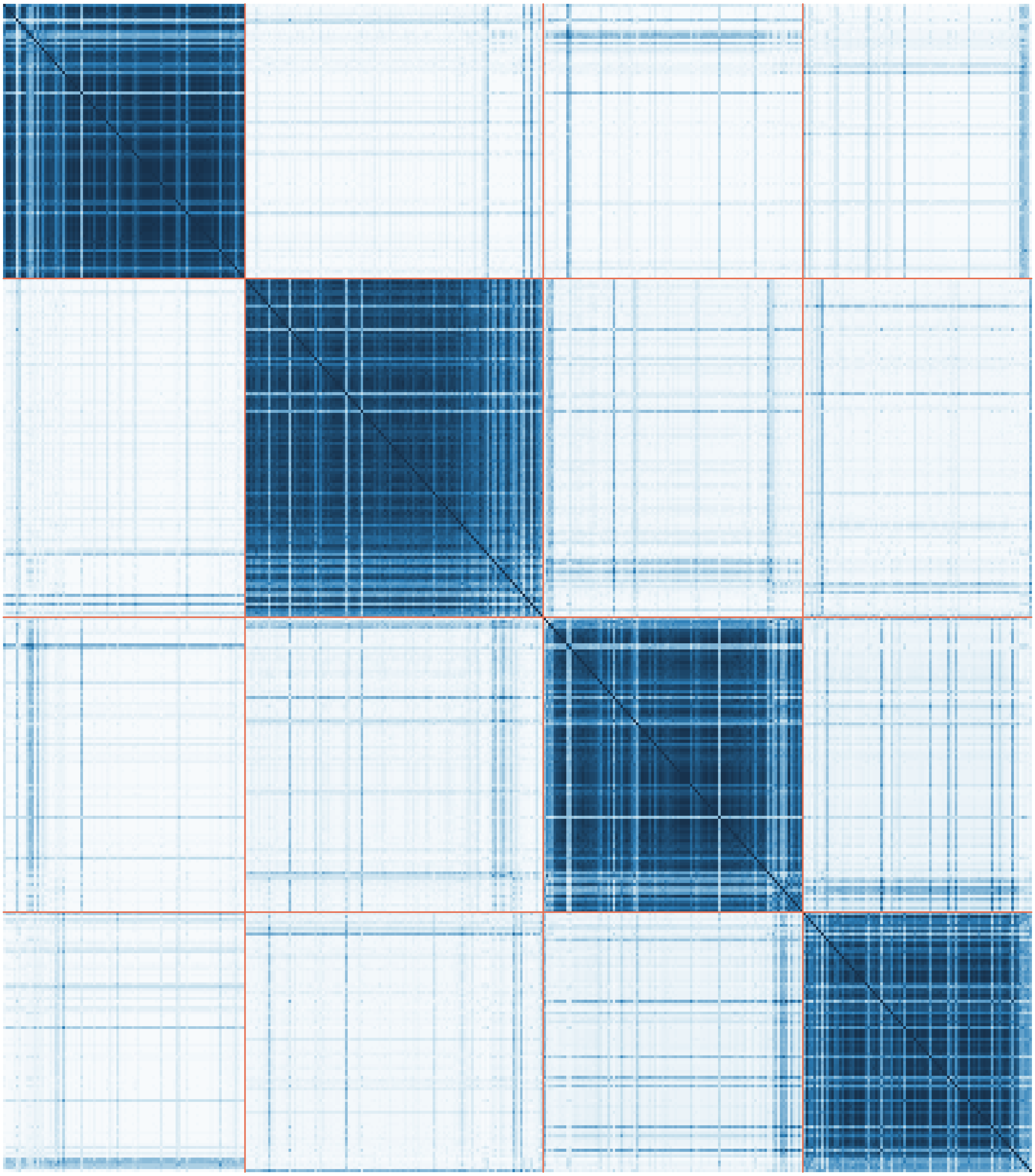
Mixed−data latent view

First two FAMD−style axes provide a common visual reference



Consensus co−assignment

Cell color is the fraction of method−imputation partitions placing two patients together



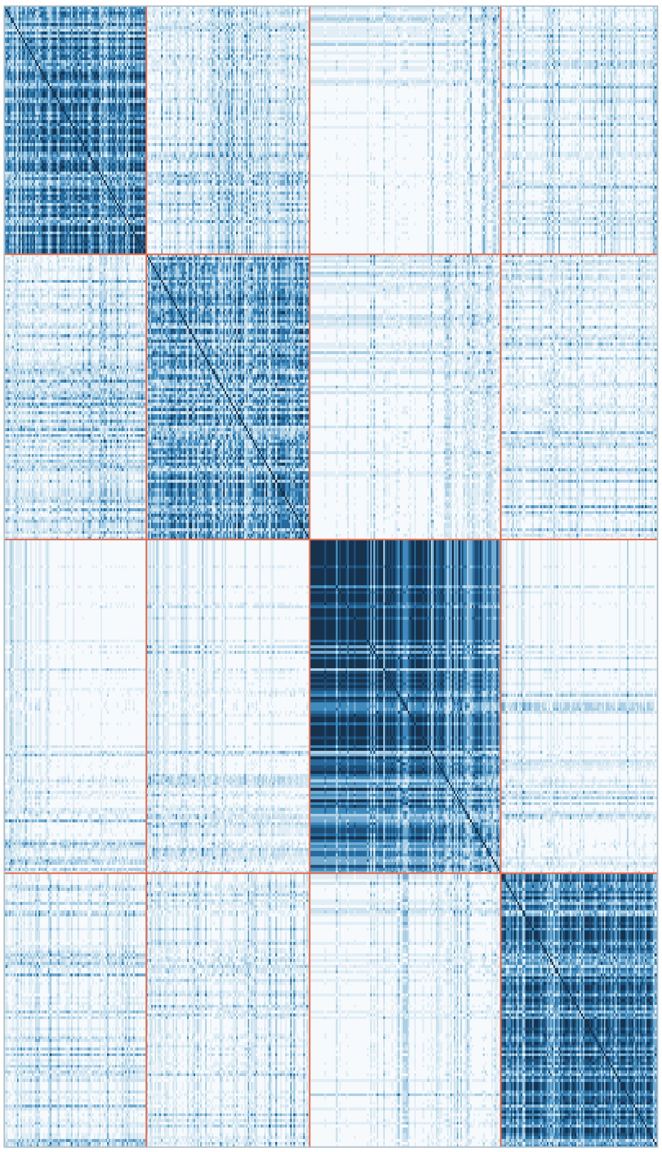
Patients sorted by consensus cluster, then supplied truth

Method heatmaps: Gower + PAM

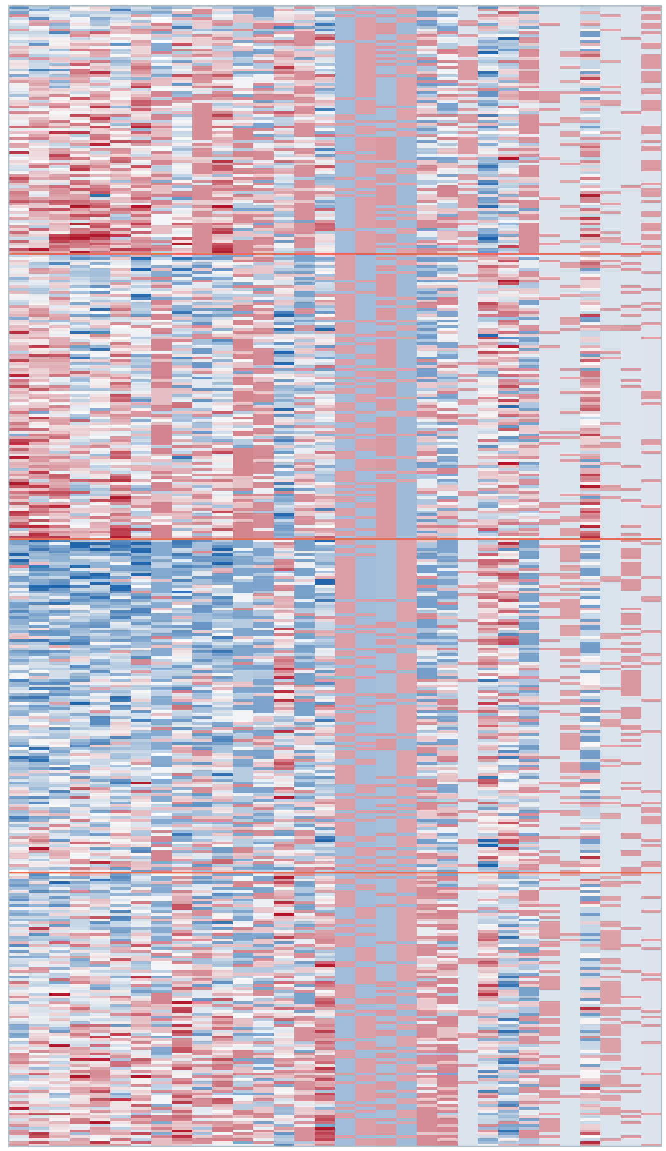
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.373	0.304	0.321	0.042
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



Cluster sizes: C1: 87 | C2: 100 | C3: 117 | C4: 96

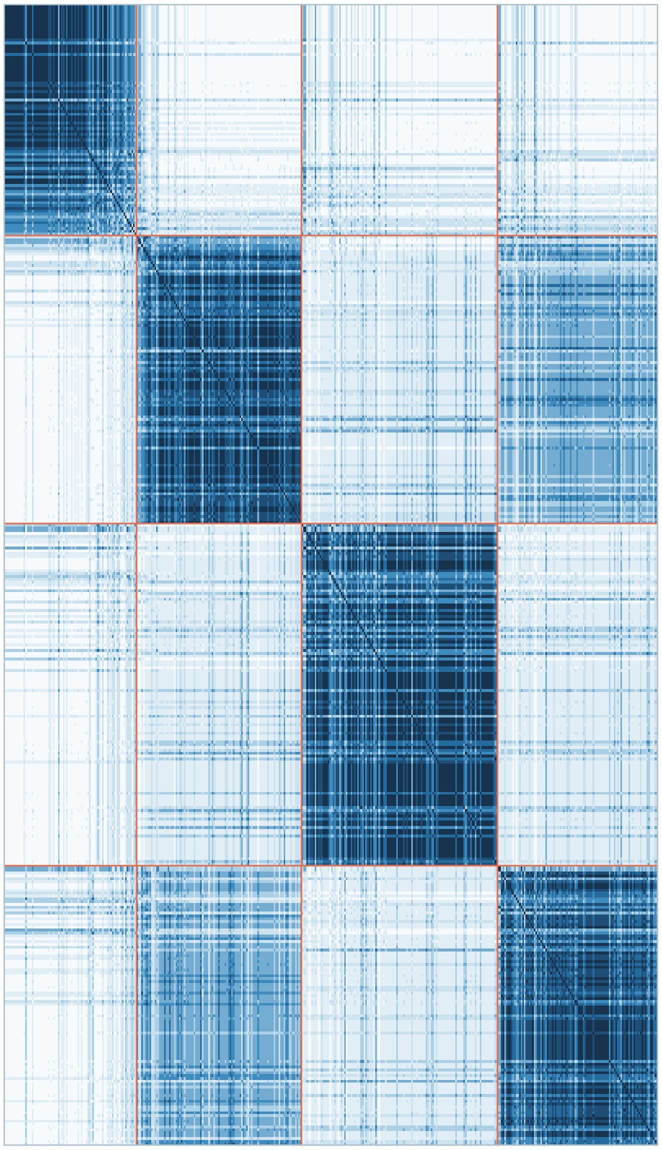
Most associated encoded columns: continuous_01, continuous_13, continuous_09, continuous_06, continuous_02, continuous_17, continuous_18, ordinal_05

Method heatmaps: Gower + hierarchical

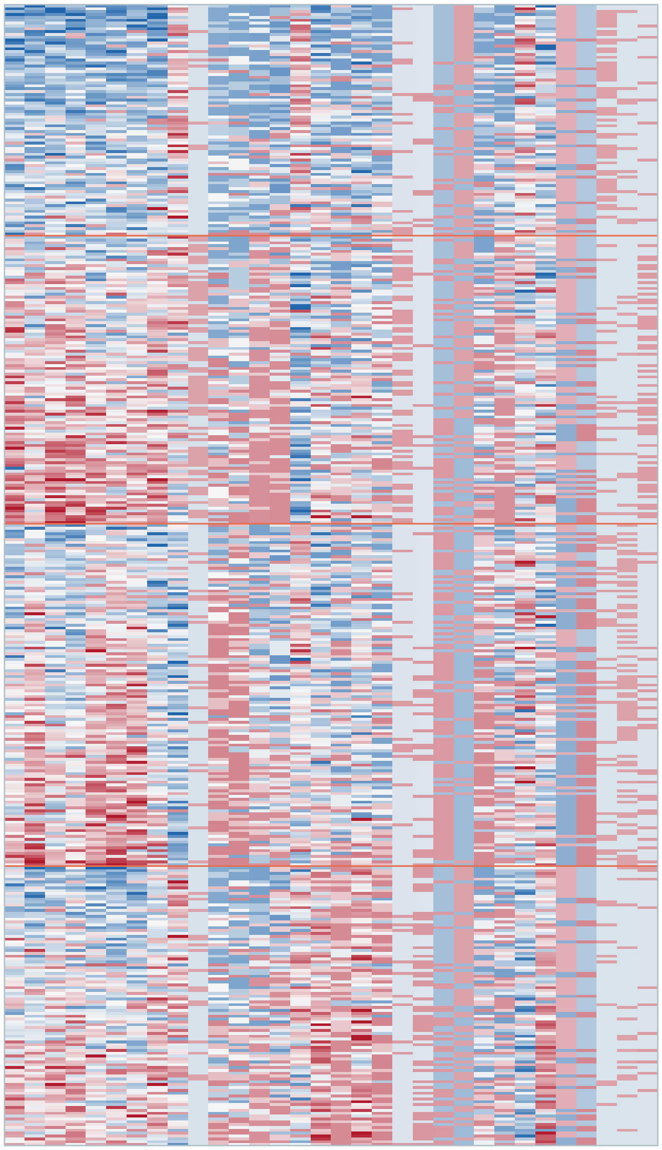
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.572	0.347	0.408	0.052
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



Cluster sizes: C1: 81 | C2: 101 | C3: 120 | C4: 98

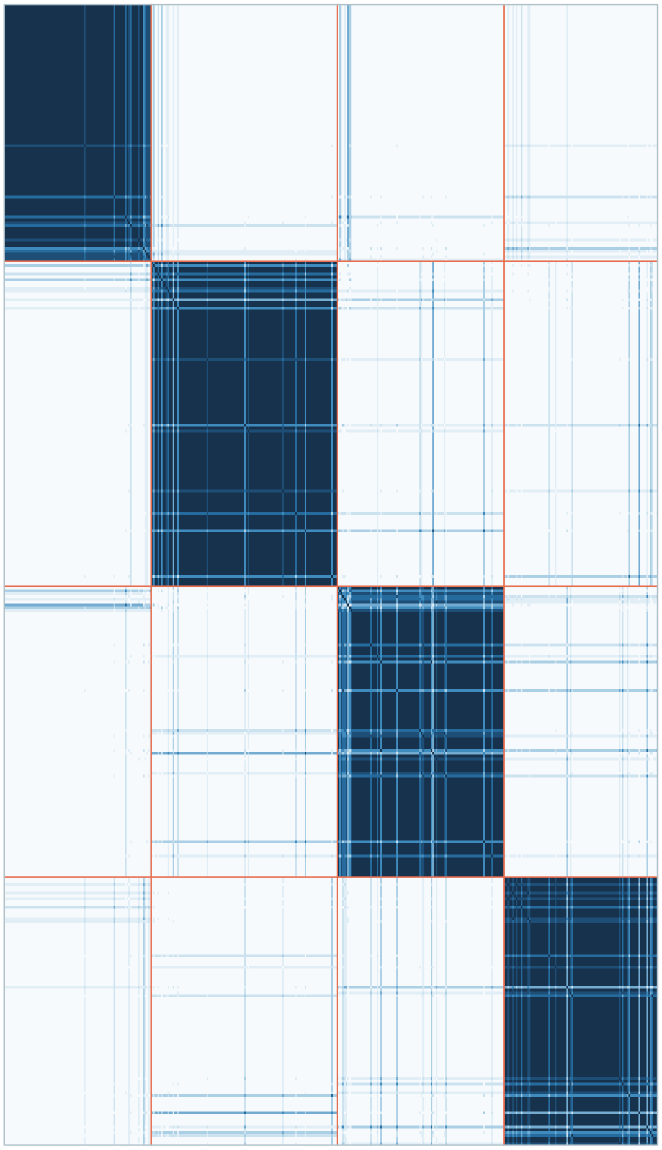
Most associated encoded columns: continuous_02, continuous_13, continuous_06, continuous_14, continuous_09, continuous_17, continuous_01, continuous_18

Method heatmaps: k−prototypes

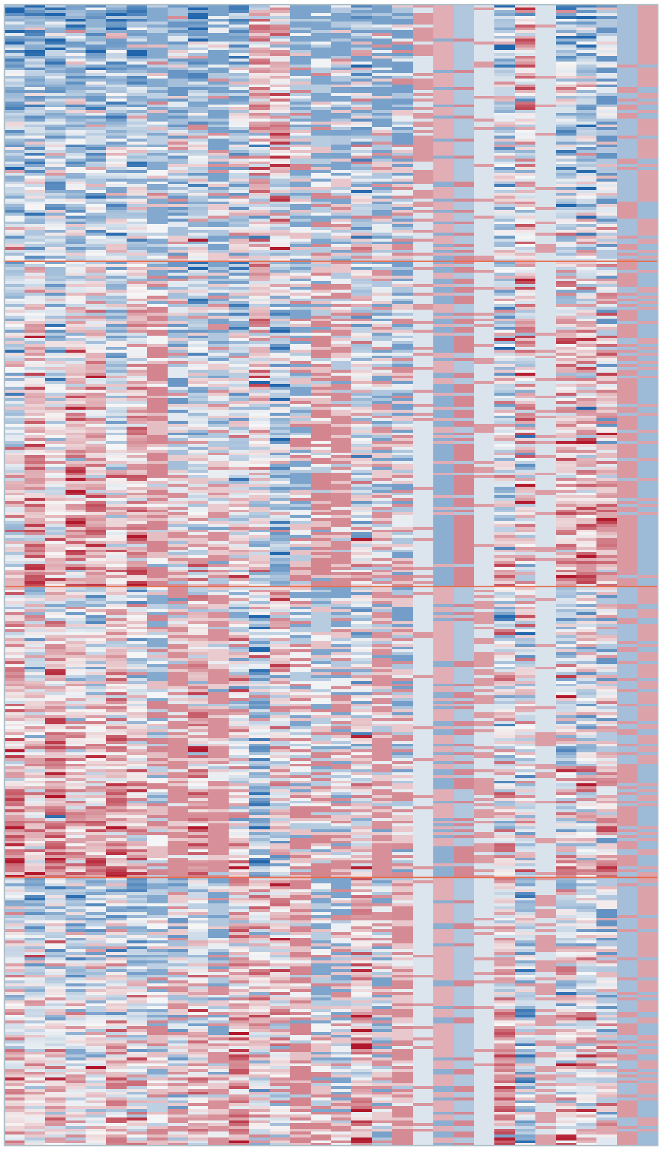
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.615	0.828	0.863	0.055
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



Cluster sizes: C1: 90 | C2: 114 | C3: 102 | C4: 94

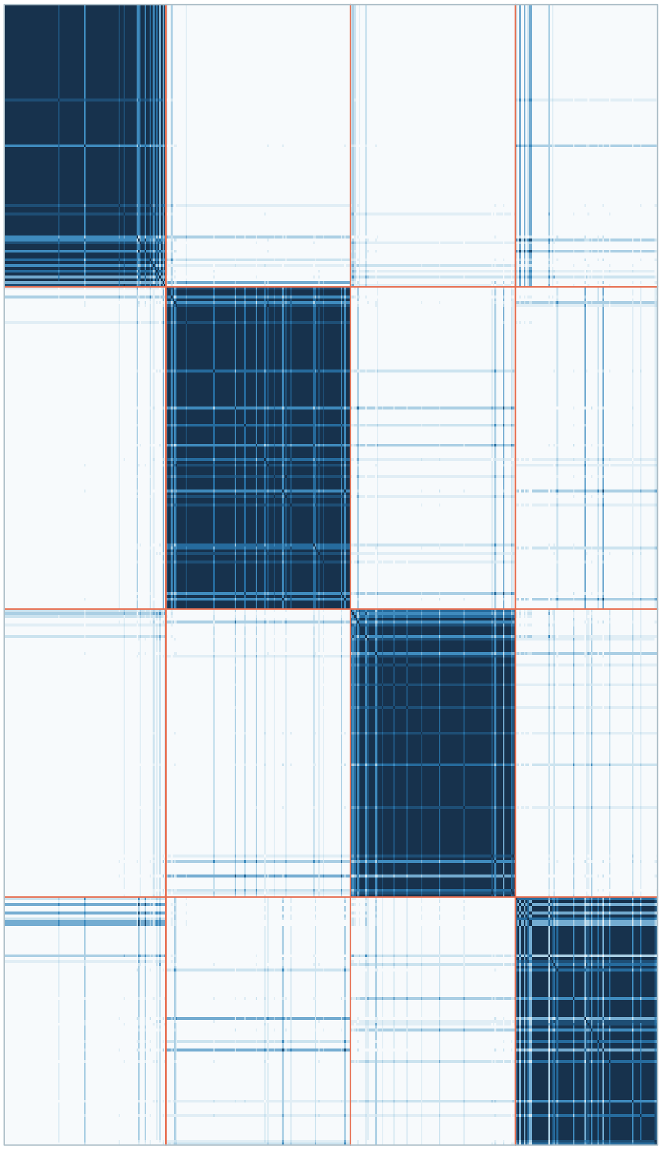
Most associated encoded columns: continuous_06, continuous_13, continuous_02, continuous_01, continuous_09, continuous_14, continuous_17, ordinal_05

Method heatmaps: FAMD−style + k−means

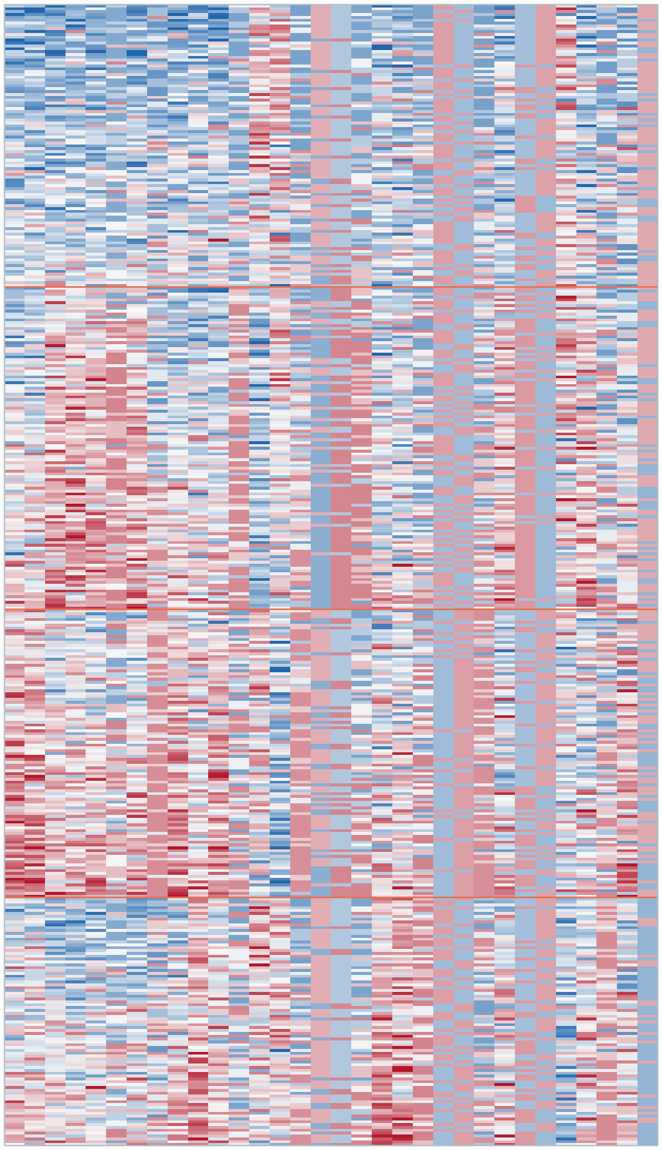
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.459	0.799	0.814	0.050
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



Cluster sizes: C1: 99 | C2: 113 | C3: 101 | C4: 87

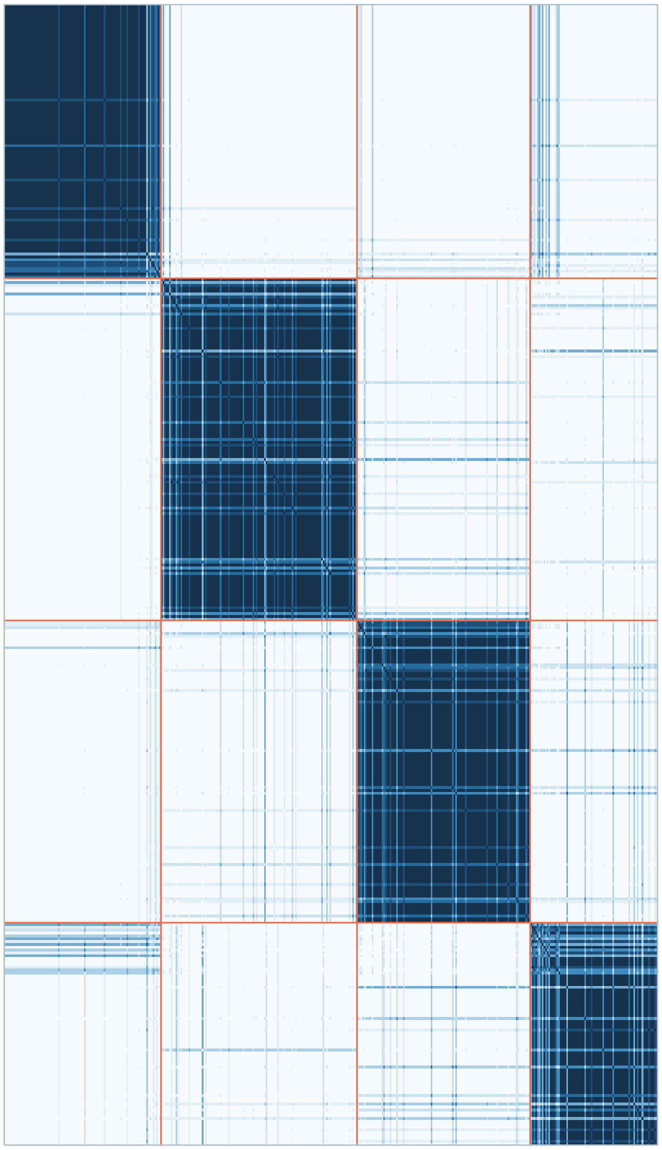
Most associated encoded columns: continuous_02, continuous_06, continuous_13, continuous_01, continuous_09, ordinal_05, continuous_17, ordinal_02

Method heatmaps: Latent−embedding mixture

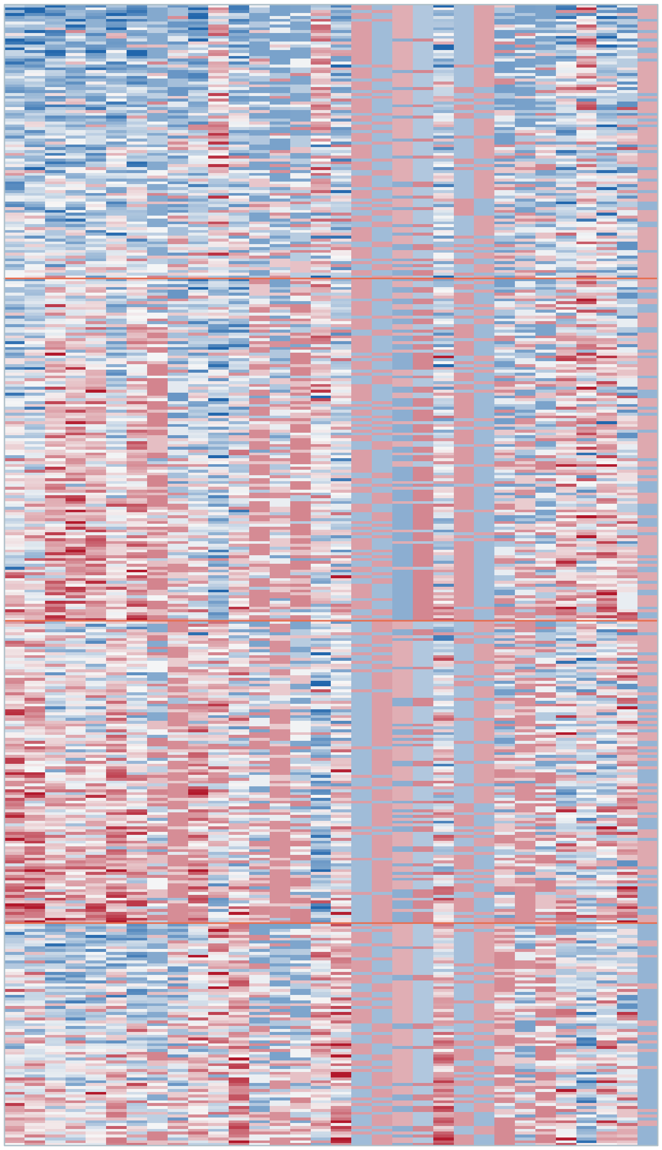
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.456	0.780	0.751	0.048
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



Cluster sizes: C1: 96 | C2: 120 | C3: 106 | C4: 78

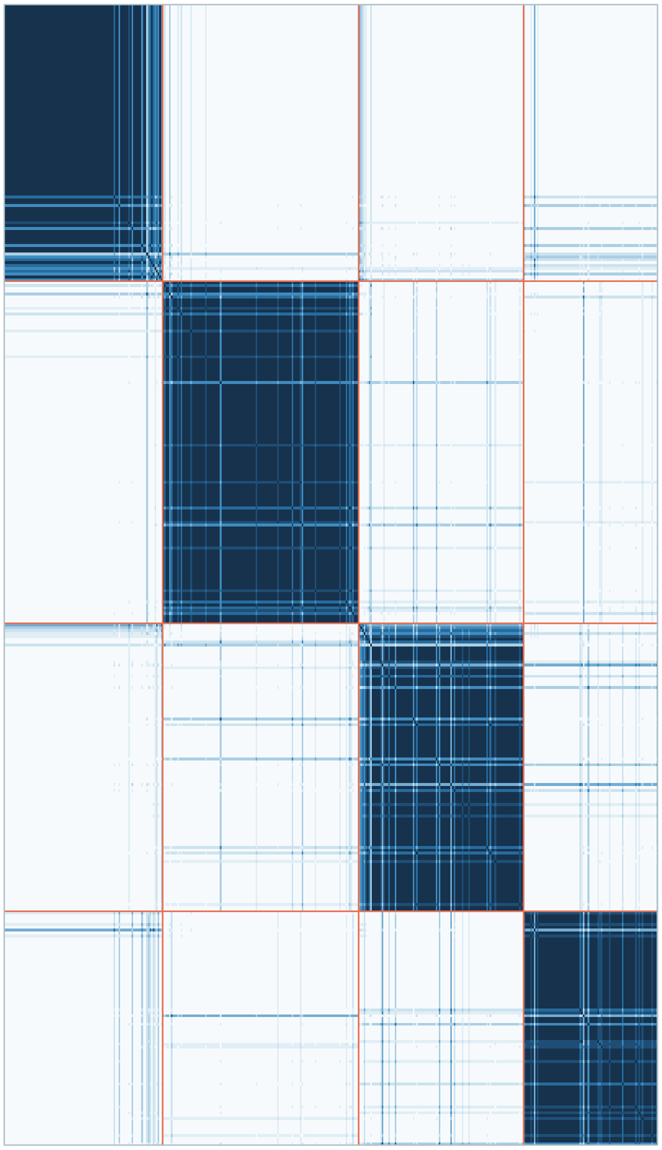
Most associated encoded columns: continuous_02, continuous_06, continuous_13, continuous_01, continuous_09, continuous_14, continuous_17, ordinal_05

Method heatmaps: Latent class/profile

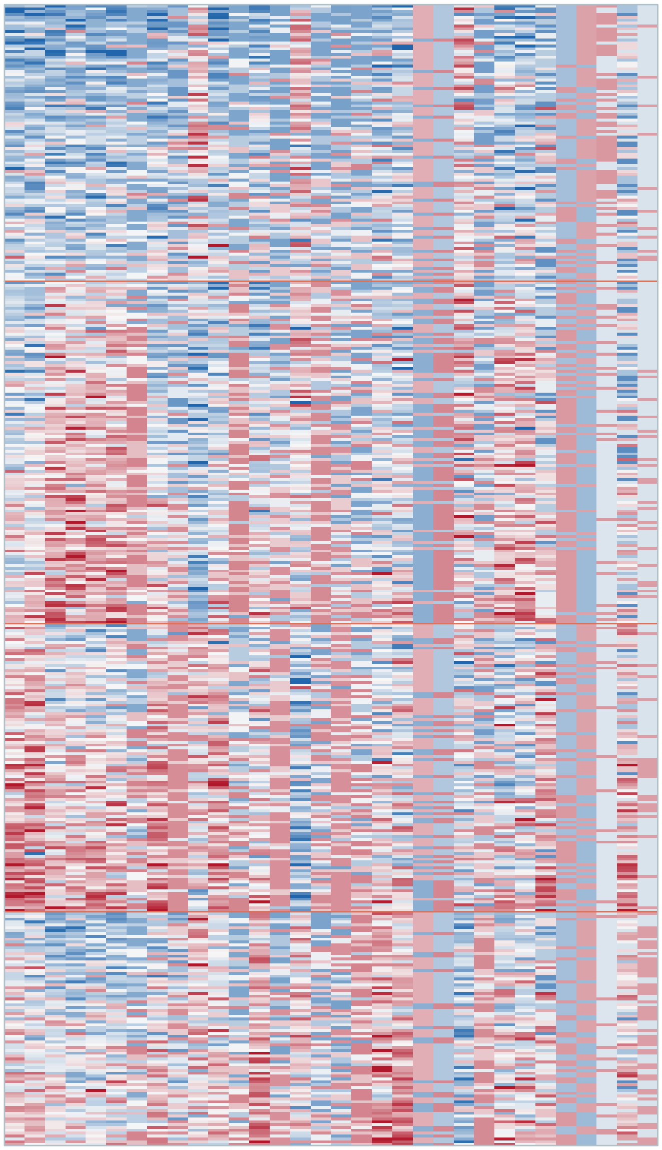
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.574	**0.813**	**0.804**	**0.052**
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



Cluster sizes: C1: 97 | C2: 120 | C3: 101 | C4: 82

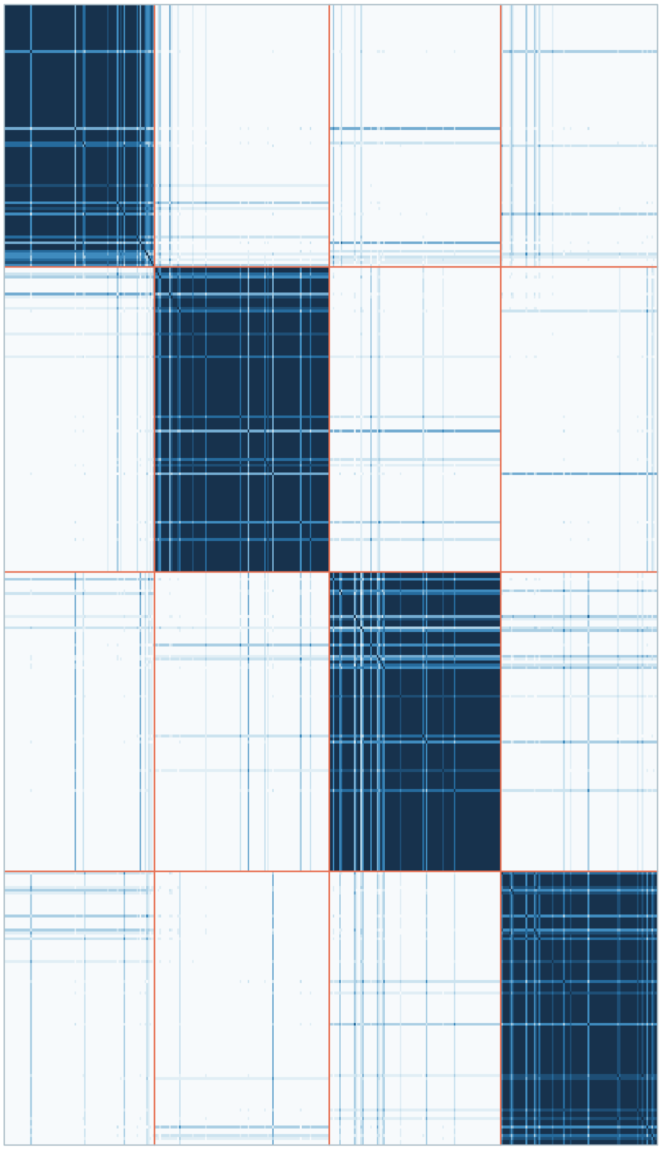
Most associated encoded columns: continuous_06, continuous_02, continuous_13, continuous_01, continuous_09, continuous_17, ordinal_05, continuous_14

Method heatmaps: Gower spectral

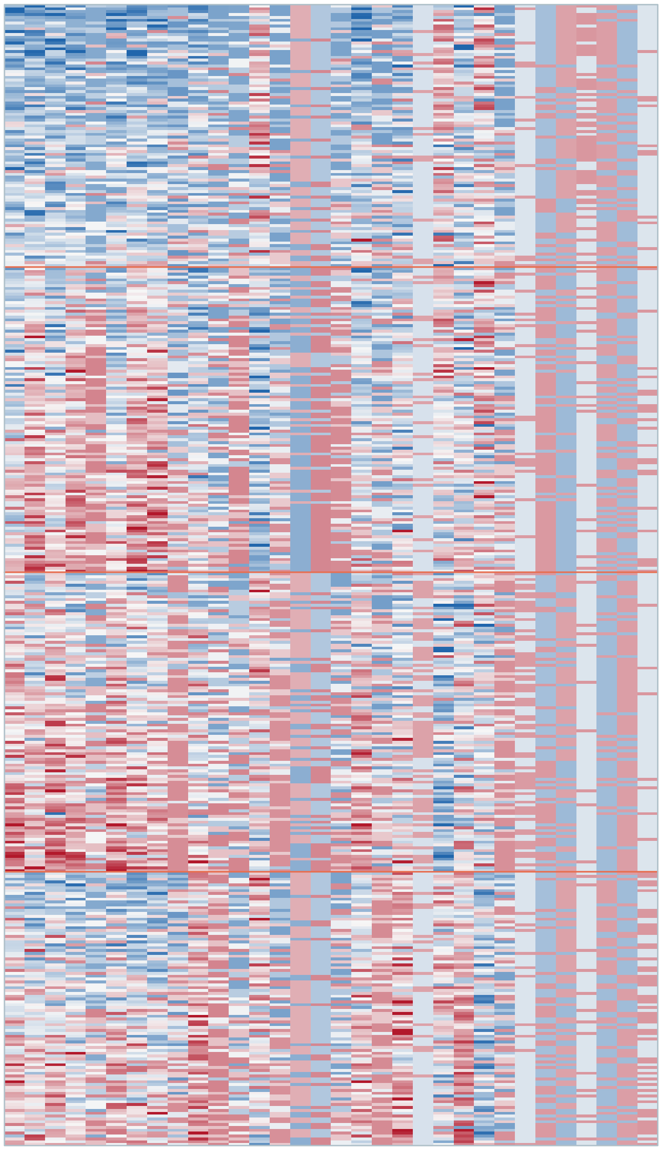
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.607	0.815	0.852	0.056
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



Cluster sizes: C1: 92 | C2: 107 | C3: 105 | C4: 96

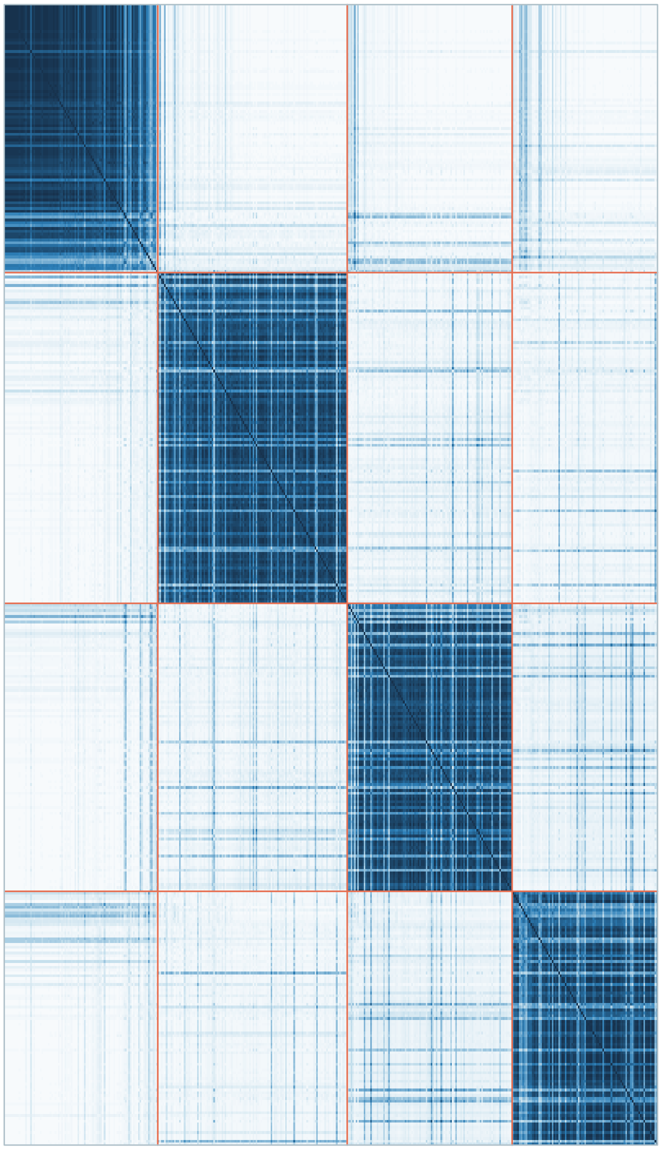
Most associated encoded columns: continuous_06, continuous_13, continuous_02, continuous_09, ordinal_05, continuous_14, continuous_17, continuous_01

Method heatmaps: Grand consensus

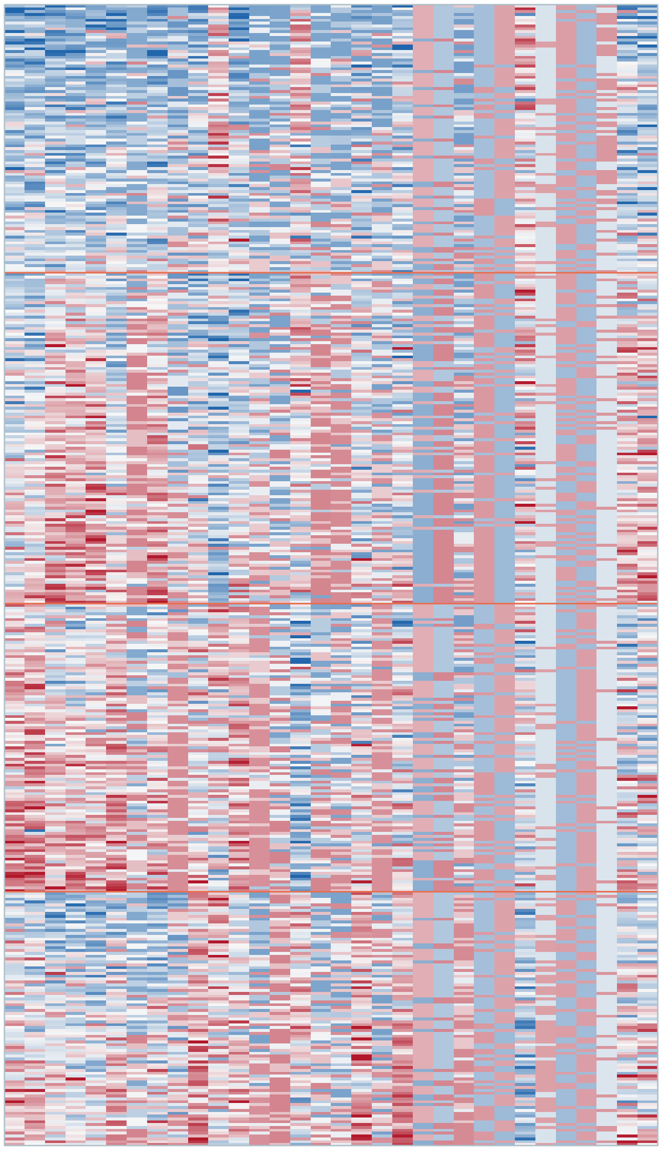
Left: stability across imputations. Right: patients by the most cluster−associated encoded features

0.554	0.794	0.838	0.054
ARI	MI stability	Subsample stability	Gower silhouette

Patient−patient co−assignment



Patient−feature heatmap (32 encoded columns)



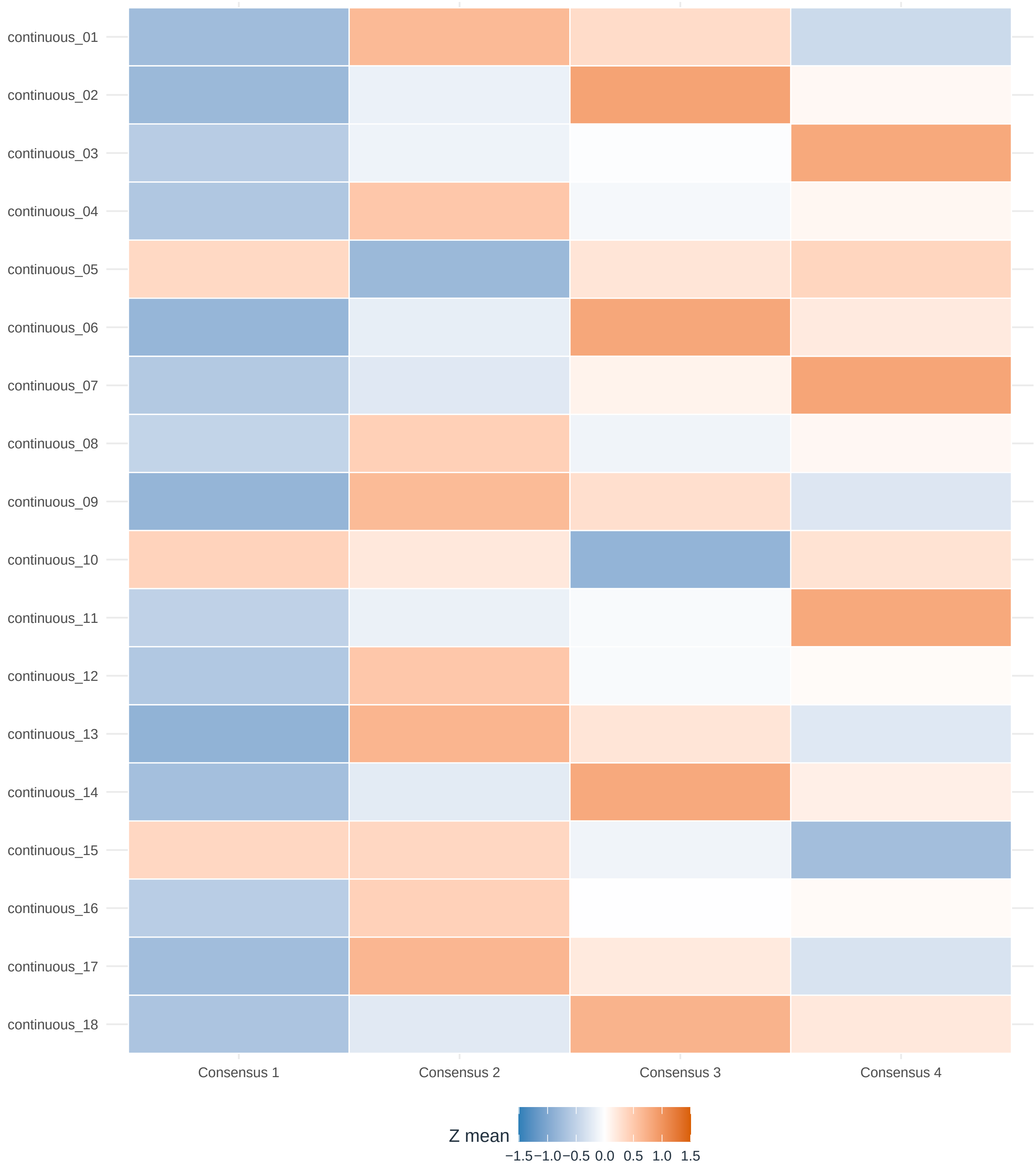
Cluster sizes: C1: 94 | C2: 116 | C3: 101 | C4: 89

Most associated encoded columns: continuous_06, continuous_02, continuous_13, continuous_09, continuous_01, continuous_14, ordinal_05, continuous_17

Consensus cluster profiles

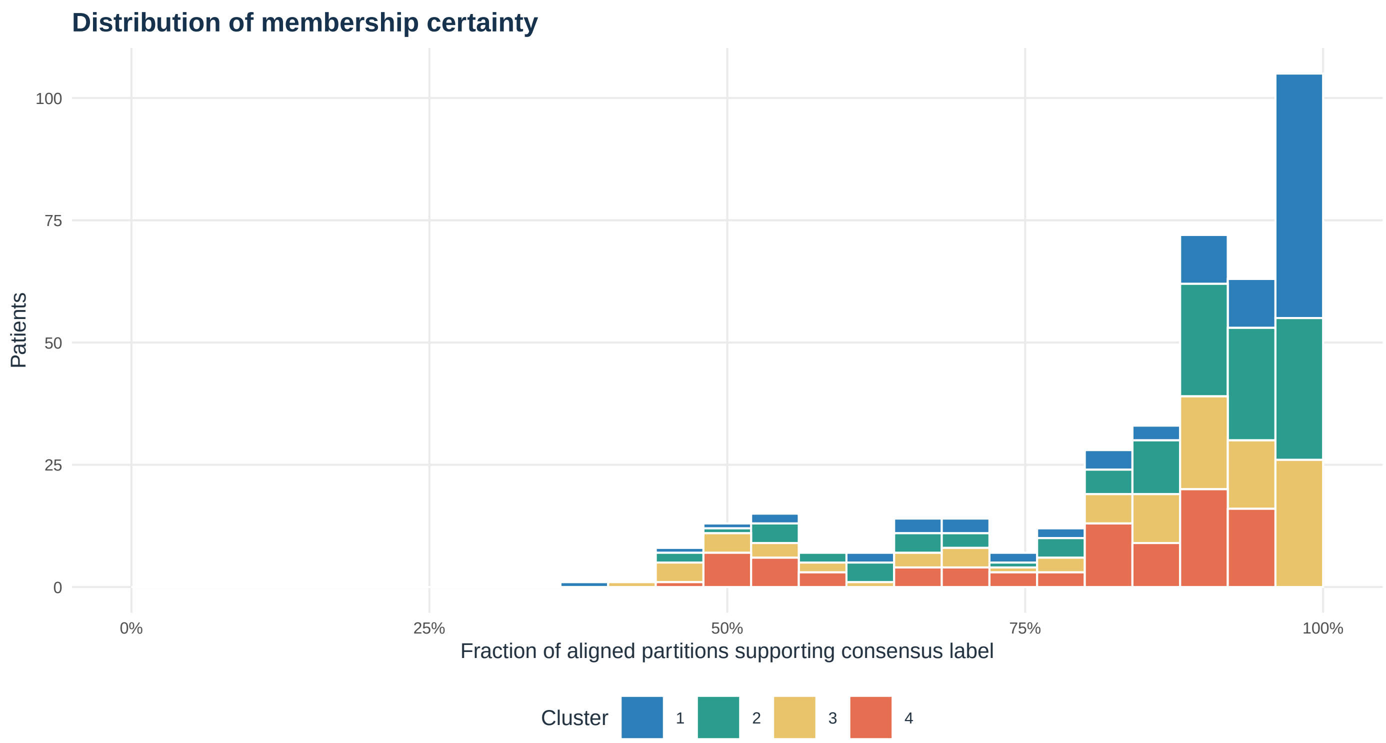
Mean standardized values for selected numeric or ordinal features

Profiles describe the fitted groups; they are not independent validation



Membership uncertainty

Consensus support exposes patients whose assignments are not robust

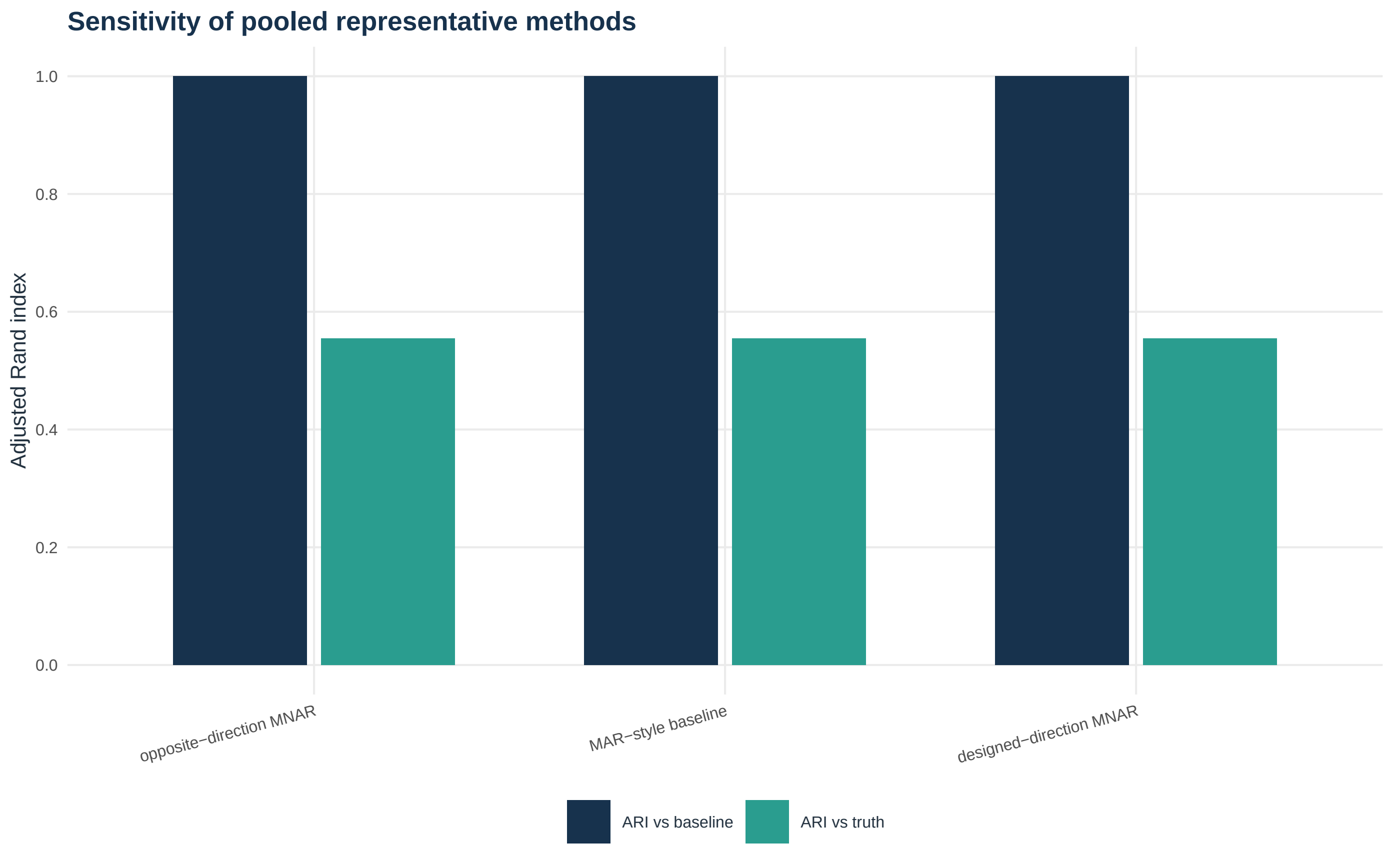


Ten least−certain patients

Patient	Truth	Consensus	Certainty
P0222	C1	1	39.3%
P0295	C2	3	42.9%
P0022	C1	3	44.6%
P0013	C2	3	46.4%
P0015	C2	3	46.4%
P0045	C4	1	46.4%
P0067	C1	2	46.4%
P0093	C1	4	46.4%
P0124	C3	3	46.4%
P0167	C1	2	46.4%

MNAR sensitivity analysis

Only originally missing values in 0 dictionary−designated features are shifted



Delta	Scenario	ARI vs truth	ARI vs baseline	Reassigned
−1	opposite−direction MNAR	0.554	1.000	0.0%
+0	MAR−style baseline	0.554	1.000	0.0%
+1	designed−direction MNAR	0.554	1.000	0.0%

No mnar_direction values were specified, so the three scenarios are identical. Edit feature_dictionary_used.csv to mark plausible directions as −1 or +1, then rerun the CSV command to obtain an informative sensitivity analysis.

How to interpret this benchmark

What transfers to a real cohort and what does not

- External recovery scores transfer only as evidence that the implementations work on this simulation. A real cohort has no hidden truth labels, and good silhouette values do not establish biological subtypes.
- The stochastic nearest−neighbor hot−deck procedure preserves mixed−data values and propagates some imputation uncertainty. For publication−grade analysis, consider a carefully specified MICE or joint−model imputation and repeat the same consensus workflow.
- The native latent class/profile model uses diagonal Gaussian distributions for continuous/log−count variables and multinomials for ordinal and nominal variables. It is intentionally parsimonious; specialized count and ordinal likelihoods may be preferable.
- The FAMD−style implementation is transparent and dependency−light but is not a drop−in reproduction of every weighting convention used by FactoMineR or PCAmixdata.
- Cluster−feature differences are descriptive because those features constructed the clusters. Validate with prespecified variables not used for clustering, then replicate in an external cohort.
- If the scientific goal is prediction, treatment selection, or outcome risk, a supervised model may be more appropriate than clustering. Clusters should not be interpreted causally.

Rerunning with your own data

The project is designed to be edited rather than treated as a black box

Fastest complete rerun

```
Rscript analyze_csv.R --input path/to/patients.csv --k auto
```

- Put one patient per row. Use −−id−column for an identifier, −−truth−column only for optional external labels, and −−exclude for outcomes, site, batch, or other columns that should not define clusters.
- A dictionary is inferred when −−dictionary is omitted and saved as feature_dictionary_used.csv. Review it, especially numeric ordinal variables, then rerun with −−dictionary pointing to the edited file.
- Allowed types are continuous, discrete, ordinal, categorical, and binary. The domain column controls block weighting. Optional role and mnar_direction columns support reporting and sensitivity analysis.
- Use −−k auto for a rank aggregation of two silhouette diagnostics and latent−mixture BIC, or provide an integer. Always inspect K−selection, stability, cluster sizes, uncertainty, and clinical coherence together.
- Increase m_imputations and n_subsamples for a final analysis. Eight and ten are intentionally modest defaults for an executable demonstration.
- Machine−readable outputs include method metrics, cluster assignments, membership certainty, missingness, K selection, MNAR sensitivity, resampling stability, the dictionary used, and a run manifest.

Project output inventory

Files produced by the reproducible pipeline

File	Purpose
feature_dictionary_used.csv	Feature types and domains actually used
auto_k_diagnostics.csv (if −−k auto)	Diagnostics used when −−k auto is selected
run_manifest.csv	Input, output, K, ID, and truth settings
results/method_metrics.csv	External, internal, stability, and runtime metrics
results/cluster_assignments.csv	Consensus and method−specific patient labels
results/membership_certainty.csv	Patient−level assignment support
results/k_selection.csv	Candidate−K diagnostics
results/mnar_sensitivity.csv	Directional departure−from−MAR scenarios
results/analysis_results.rds	Complete report−ready analysis object
output/pdf/*_clustering_report.pdf	This rendered benchmark report

Machine−readable files are written below the chosen −−output−dir. PDF reports are always written to output/pdf; −−pdf changes only the filename.

Reproducibility record

Software information captured during execution

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R version 4.6.1 (2026−06−24)
Platform: aarch64−apple−darwin23
Running under: macOS Tahoe 26.6.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1

locale:
[1] C.UTF−8/C.UTF−8/C.UTF−8/C/C.UTF−8/C.UTF−8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
[1] compiler_4.6.1  cluster_2.1.8.2
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Analysis timestamp (UTC): 2026−09−03 02:08:01